Jeff Shen

Statement of Interest

I am excited to be applying for this incredible opportunity at the Institute for Research on Exoplanets. My fascination with space has always been because of the humility it inspires in me—I find it absolutely incredible that I am alive at this moment, despite life being so (as far as we know) unlikely. The research focus of iREx is appealing to me because the study of exoplanets and planetary systems—apart from being an exciting and promising area of research—is intricately linked to the discussion of extraterrestrial life and some of the fundamental questions that I am interested in exploring. What conditions observed in our Solar System are unique? How and where are (Earth-like) planets formed, and what are they and their atmospheres composed of? How does the evolution of the host star influence such things? And of course, the big question: are we alone? These topics are interesting in themselves; moreover, answers to these questions will not only provide great implications for the origins and evolution of our own Solar System, but also contribute to our greater understanding of the universe. I hope to immerse myself in the exploration of these areas, and I see no better place to do that than at iREx.

From my understanding, the internship will likely involve a significant amount of coding and statistical analysis. The explosion of data and its impact on astronomy are apparent, and being able to understand this information is critical. I have independently sought to develop the skills necessary to do so—as a first-generation student, being able to tackle new problems independently is not foreign to me. I have years of experience programming in Python, with a focus on scientific computing using packages like numpy, pandas, matplotlib, and astropy. I have also learned to leverage the power and versatility of machine learning and deep learning techniques through independent coursework, and have refined those abilities by applying them in independent projects. My broad programming skillset—which includes shell scripting, SQL, and cloud computing—allows me to tackle a wide variety of problems, and my academic background in statistics has equipped me with the tools necessary to do so rigorously. Perhaps equally as important is being able to communicate, which I believe to be critical to the scientific research process. My experience teaching programming classes, and the numerous report-based projects that I have completed—including a research report I wrote on mass-loss mechanisms in late-stage AGB stars for an astronomy course on stars and planets—have honed my ability to communicate effectively, both verbally and through writing. I am confident in my ability to convey information concisely. These qualities position me as a strong candidate to pursue research in this area.

The most attractive aspect of this internship to me is the unique atmosphere. Reading about past interns' experiences with the program has led me to understand that the nature of this program is like no other. It brings together the foremost researchers in the field of exoplanets, and as a student with much to learn, I would cherish the experience that a mentor could provide. In particular, I find Professor Doyon and researcher Artigau's involvement with the SPIRou spectrograph to be appealing. The instrument's focus on the detection of habitable Earth-like planets aligns strongly with my own research interests, and being able to work so closely with those who are most familiar with it would be a boon. Furthermore, the Institute plays a pivotal role in the development of immensely impactful instrumentation, such as the JWST NIRISS, and being able to work with cutting-edge tools and data is thrilling. The autonomy offered by the program will also allow me to tackle problems creatively. Most of all, I am sure that the program will attract some of the brightest and most curious minds; I would love nothing more than to be immersed in an environment where I can contribute to the ongoing discourse on a topic that I, and all others present, are passionate about. I believe that iREx is unique in that it can offer me the best opportunity to do so.

I firmly believe that my research interests align closely with those of the Institute. Furthermore, my ability to program, communicate effectively, and work independently, together with my curiosity and passion, would make a positive contribution to the team. This internship program offers a truly unique opportunity to conduct research in the exciting field of exoplanets—I hope to learn from the best and to work with others who are as excited about this area as I am. I sincerely thank you for your time and consideration.